Bilinear pairings are used to prove that product of two no's is equal to a 3rd number by encrypting (not revealing) any of the three numbers.

$\mathcal{E}(a) \cdot \mathcal{E}(b) = \mathcal{E}(c)$; $\mathcal{E} \rightarrow$ Encryption function

- Symmetric and Assymmetric groups:

$e(aG, bG)$ / $e(aG_1, bG_2)$

Symmetric

Groups having same no. of points

Assymmetric

Properties of bilinear pairing :

$$e(aG_1, bG_2) = e(G_1, abG_2)$$
$$= e(abG_1, G_2)$$

- with regards to Ethereum, G_2 is field extension of G_1, i.e. each element of G_2 is two dimensional so each element in G_2 has 4 coordinates (x_1, x_2, y_1, y_2)

What does $e(a,b)$ output ?

It outputs an element of another group (G_T).

The group G_T has identity element

$$\underbrace{(1, 0, 0, - - -)}_{12 \text{ elements}}$$

All elements of G_T group are 12 dimensional tuples.

And G_T group is a finite cyclic group.

NOTE: Elements in G_T behave like powers of a base.

i.e. $e(3G_2, 2G_1) = b^6 G_T$

$3 * 2$

- EIP 197:

Adds precompiles to enable verification of ZK proofs using bilinear pairings

address of contract : 0x8

Q If a group is cyclic and finite, can we always construct a homomorphism to any other finite cyclic group from the previous group?

A YES!